AHFS-TA: Adaptive Hierarchical Feature Selection with Temporal Attention for Student Dropout Prediction Using Machine Learning and Deep Learning Approaches

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Abstract

Student dropout remains a critical challenge in higher education, affecting institutional success metrics and student welfare. Accurate early prediction enables timely intervention, yet existing approaches struggle with temporal dynamics and feature interpretability. This paper introduces AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework that synergistically combines hierarchical feature selection, temporal attention mechanisms, and explainable AI for comprehensive dropout prediction. Our methodology employs five feature ranking methods (Information Gain, Gain Ratio, Gini Index, Chi-squared, and ANOVA F-statistic) to identify influential predictors across 34 unique features spanning academic, financial, and demographic categories. We conduct extensive experiments on a real-world educational dataset comprising 4,424 students with three outcome classes (Dropout: 1,421, Enrolled: 794, Graduate: 2,209). Comparative analysis against six baseline models—Decision Tree (67.0%), Naive Bayes (70.9%), Random Forest (76.7%), AdaBoost (74.2%), XGBoost (75.9%), and Neural Network (71.4%)—demonstrates that AHFS-TA achieves state-of-the-art performance with 91.32% accuracy and 95.5% AUC-ROC. Furthermore, we integrate SHAP-based explainability for all models, revealing that curricular unit performance, tuition fee status, and scholarship possession are the most influential dropout predictors. Our framework addresses the complete machine learning pipeline from data preprocessing to model deployment, providing actionable insights for educational administrators through interpretable visualizations.

Index Terms

Student dropout prediction, machine learning, deep learning, feature selection, temporal attention, explainable AI, SHAP, higher education, educational data mining

I. Introduction

St dropout in higher education institutions represents a multifaceted problem with far-reaching consequences for students, institutions, and society at large. When students discontinue their academic pursuits before completing their degrees, they face diminished career prospects, accumulated student debt without corresponding credentials, and psychological impacts from perceived failure [1], [2]. Institutions suffer from reduced graduation rates, lost tuition revenue, and reputational damage that affects future enrollment. From a societal perspective, dropout wastes educational resources and diminishes the skilled workforce pipeline essential for economic development [3], [4].

The magnitude of this challenge is substantial. Global statistics indicate that approximately one-third of undergraduate students fail to complete theirr programs within the expected timeframe [5].In developing nations, high tertiary dropout rates remain a critical challenge, with completion rates consistently and significantly lower than in high-income countries, creating major barriers to social mobility and economic advancement (UNESCO, 2022; World Bank, 2023). These statistics underscore the urgent need for predictive systems that can identify at-risk students early enough for effective intervention.

Traditional approaches to dropout prediction have evolved from simple demographic analysis to sophisticated machine learning models. Early studies focused on individual risk factors such as socioeconomic status, prior academic performance, and first-year grades [6], [7]. While informative, these approaches often missed the dynamic, temporal nature of student disengagement—a gradual process influenced by evolving circumstances rather than static characteristics [8].

The advent of educational data mining and learning analytics has enabled more nuanced predictive modeling [9], [10]. Machine learning algorithms can process multidimensional student data to identify complex patterns invisible to traditional statistical methods [11], [12]. However, current approaches face three critical limitations that our research addresses:

Challenge 1: Feature Selection Complexity. Educational datasets contain dozens of features spanning academic performance, financial status, demographic background, and behavioral patterns. Existing methods typically employ single feature selection techniques, potentially missing important predictors that rank highly under alternative criteria [13]. Our framework addresses this through multi-method feature ranking combining five established techniques.

Challenge 2: Temporal Dynamics Neglect. Student engagement evolves across semesters, with dropout decisions often following semester-specific crises (failed courses, financial difficulties, personal issues). Most prediction models treat student data as static snapshots, ignoring the temporal patterns that signal declining engagement [14]. Our AHFS-TA framework incorporates temporal attention mechanisms that weight semester-specific contributions.

Challenge 3: Explainability Gap. While ensemble methods and deep learning achieve high accuracy, their "black-box" nature limits practical utility. Educational administrators need to understand *why* a student is flagged as at-risk to design appropriate interventions [15]–[17]. Our comprehensive SHAP integration provides interpretable explanations for all models.

This paper makes the following contributions:

1) **Novel Framework**: We introduce AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), integrating LLM-based semantic feature enrichment through DistilBERT, bidirectional GRU with multi-head attention, and three-stream adaptive feature selection.
2) **Comprehensive Baseline Comparison**: We evaluate six baseline models (Decision Tree, Naive Bayes, Random Forest, AdaBoost, XGBoost, Neural Network) using identical preprocessing and evaluation protocols, establishing fair benchmarks.
3) **Multi-Method Feature Ranking**: We apply five feature ranking techniques (Information Gain, Gain Ratio, Gini Index, Chi-squared, ANOVA F-statistic) with meta-ranking aggregation to identify robust dropout predictors.
4) **Explainable AI Integration**: We implement SHAP analysis for all six baseline models plus AHFS-TA, providing transparent feature importance explanations.
5) **Reproducible Evaluation**: We report complete metrics including accuracy, precision, recall, F1-score, confusion matrices, ROC curves, AUC scores, and 10-fold cross-validation with standard deviations.

The remainder of this paper is organized as follows: Section II reviews related work in dropout prediction. Section III describes the dataset and preprocessing. Section IV details feature ranking methodology. Section V presents baseline models. Section VI introduces the proposed AHFS-TA framework. Section VII describes experimental setup and evaluation metrics. Section VIII presents results and comparative analysis. Section IX discusses explainability findings. Section X concludes with future directions.

II. Related Work

Student dropout is a global concern because it increases the risks of crime, social issues, hinders overall societal progress, and also leads to poverty. Dropping out not only hinders students' personal and professional development but also leads to institutional resource losses and broader socio-economic challenges. When researchers identify the underlying issues such as poverty or a lack of resources, along with other factors using educational data mining (EDM), they can develop effective prediction and prevention strategies using AI, Machine Learning and Deep Learning methods.

Early dropout prediction relied on logistic regression and discriminant analysis to identify risk factors [15], [18]. Tinto's integration model [19] emphasized the interplay between academic and social integration, inspiring subsequent research to incorporate engagement metrics. While interpretable, these methods assume linear relationships and struggle with high-dimensional feature spaces. Recent comprehensive reviews have highlighted the limitations of traditional approaches when dealing with the complexity of modern educational datasets [20], [21].

With deep learning and generative AI approaches, we can learn hierarchical representations from raw data, and also make a significant advancement in predictive modeling for student dropout. We can build robust early-warning systems by integrating advanced AI methods with educational data that can support educational institutions in reducing dropout rates, providing explanations of the root causes of dropping out, intervention strategies, and improving overall student success.

From studying different research on student dropout, we found that academic performance is the strongest predictor of dropout rates. Delogu et al. [22] identified that the number of ECTS credits achieved in the first year serves as the most significant warning signal. They utilized an extensive national administrative dataset (n=144,904). Vaarma et al. [23] emphasized the importance of LMS data and found that grades and earned credits were the "most crucial" factors. Likewise, Schnedlitz et al. [24] showed that direct assessments scores (such as score of exams and assignments) were the most influential indicators of eventual outcomes. From their studies, we found that inadequate academic performance, particularly in the initial stages of a student's career, is a key sign of drop out risk.

We know that grades are important, but they frequently serve as a delayed measure. From Studies, we found that the behavioral data can offer significant early indications. Vaarma et al. [23] discovered that digital engagement, such as the number of logins and clicks (metrics from a Learning Management System (LMS)), ranked as a "very important" predictor, coming in third after academic metrics. Schnedlitz et al. [24] did an in-depth analysis on submission patterns in a computer science class. They illustrated that performance in lab exercises and submission patterns could forecast dropouts with an accuracy of 80

A student's background is an important factor in predicting dropout. It plays a crucial role in their ability to persist in their studies. Delogu et al. [22] identified that family income has an impact on student dropout. They found that the cost of tuition and the distance from home to university were significant factors. They discovered that these factors are a stronger

TABLE I: Comparison with Recent Literature

Study	Dataset	Accuracy/AUC	Temporal	Multimodal	LLM/XAI
Basnet et al. (2022) [29]	200,904 samples	87.64%	No	Yes	No
Phan et al. (2023) [30]	14,391 students	80%	No	Text + Tabular	No
Krüger et al. (2023) [31]	299,722 entries	38.22% – 89.50%	Cumulative and Delta Features	Yes	SHAP
Leelaluk et al. (2024) [32]	490 students	82%	GRU + Attention (Attn-ANN)	No	Attention Weights (XAI)
Chaikalis et al. (2024) [33]	790 students	80.46%	No	No	No
Padmasiri et al. (2024) [25]	4424 students	80%	No	No	SHAP + LIME
Klinke et al. (2024) [34]	5,796 students	70.6%	No	No	No
Vaarma et al. (2024) [23]	8,813 students	85.3%	Yes	Yes	No
Diaz et al. (2024) [35]	288 students	91%	No	No	No
Okoye et al. (2024) [36]	52,262 students	90.9%	No	No	No
Okoye et al. (2024) [36]	53,639 students	81.7%	No	No	No
Shou et al. (2025) [37]	4918 students	82%	LSTM	Video + Tabular	No
This Work AHFS-TA	**4,424**	**91.32%**	**BiGRU + Attn**	**DistilBERT + Tabular**	**SHAP**

predictor in the more affluent northern areas. Similarly, Padmasiri et al. [25] confirmed that financial conditions (such as whether tuition payments were made on time) ranked among the most important predictors. They used data from a public Kaggle dataset. Villar et al. [26] found an evidence that dropout rates are not merely an academic problem, but are closely linked to a student's financial situation and social support network. They found this evidence after incorporating socioeconomic aspects like scholarship status and the occupation of parents into their effective predictive model.

Psychological theories are very important. We need it to understand the mind beyond observable behavior. Breetzke et al. [27] study provides a detailed perspective using Expectancy-Value, and they discovered that students face the greatest risk when they possess both a lack of confidence in their potential success (expectancy) and perceive little value or significant cost associated with their education (value). They focused on understanding the motivational factors to boost student's motivation and academic assistance. They found that enhancing perceived value can offset low confidence.

In previous days, conventional statistics were used, but nowadays, it has evolved to advanced machine learning models. Goran et al. [26] discovered that boosting techniques (LightGBM, CatBoost) surpassed traditional methods and conducted a comparison of algorithms. Rabelo et al. [28] (Brazil) found 89

From the literature review, we learned that machine learning models are increasingly providing high levels of predictive accuracy in identifying student dropout, using several factors such as academic performance, psychological elements, socioeconomic status, and engagement. However, most of the current research is conducted with conventional machine learning techniques or lacks a comprehensive approach that combines various types of information, including demographic, academic, and socioeconomic characteristics. Although most of the research finds the important risk factors, very few provide tailored, actionable, and interpretable insights that both institutions and students can practically utilize.

Table I positions this work within the landscape of recent educational prediction research, highlighting the unique contributions of our proposed approaches. As demonstrated in Table I, our AHFS-TA framework achieves state-of-the-art performance (91.32% accuracy) by uniquely combining temporal attention mechanisms with multimodal feature fusion and LLM-based explainability. While previous studies have explored individual components (temporal modeling [38], [39], multimodal learning [40], [41], explainability [42], [43]), our work is the first to integrate all three dimensions: sophisticated temporal modeling via bidirectional GRU with multi-head attention, multimodal fusion of structured tabular data with DistilBERT-extracted psychosocial features, and comprehensive model-agnostic explainability through SHAP (SHapley Additive exPlanations) analysis.

Despite significant advances in recent literature [31], [42], [44], several critical gaps persist: We have found limited integration of temporal attention with adaptive feature selection and lack of unified explainability frameworks for multiple classifiers. Another limitation we found that underutilization of multi-method feature ranking consensus with limited exploration of LLM-enhanced feature extraction for educational data.

Our AHFS-TA framework directly addresses these gaps through its novel architecture and comprehensive evaluation methodology.

III. Dataset Description

A. Dataset Overview

We utilize a real-world educational dataset from a higher education institution, comprising comprehensive student records suitable for dropout prediction research.

The three-class formulation reflects real-world complexity: distinguishing students who graduate, those who dropout, and those still enrolled (requiring different intervention strategies).

This three-class formulation presents unique challenges compared to binary dropout prediction. The "Enrolled" class represents students who have neither graduated nor dropped out—they may be continuing their studies, on leave, or in transition.

TABLE II: Dataset Statistics

Attribute	Value
Total Students	4,424
Total Features (Original)	46
Total Features (After Preprocessing)	34
Number of Classes	3
Class Distribution	
Dropout	1,421 (32.1%)
Enrolled (Continuing)	794 (17.9%)
Graduate	2,209 (50.0%)
Train/Test Split	80%/20%
Training Samples	3,539
Test Samples	885

This intermediate class is inherently ambiguous, as these students could eventually move to either outcome. From a practical standpoint, correctly identifying enrolled students is crucial for targeted retention interventions, as they represent the population most amenable to influence.

The class distribution exhibits moderate imbalance, with the Graduate class comprising half of all instances. This imbalance necessitates careful handling during model training to prevent bias toward the majority class. We address this through stratified sampling, class-weighted loss functions, and balanced performance metrics.

B. Feature Categorization

Features are organized into three semantic categories with some overlap:

1) Academic Features (18 Features): Academic features capture curricular engagement across two semesters:

- **Curricular Units 1st Semester**: Credited, Enrolled, Evaluations, Approved, Grade, Without Evaluations (6 features)
- **Curricular Units 2nd Semester**: Credited, Enrolled, Evaluations, Approved, Grade, Without Evaluations (6 features)
- **Academic History**: Previous qualification grade, Admission grade, Application mode, Application order, Course, Daytime/evening attendance (6 features)

2) Financial Features (12 Features): Financial features capture economic circumstances:

- **Direct Financial**: Tuition fees up to date, Scholarship holder, Debtor
- **Macroeconomic Indicators**: Unemployment rate, Inflation rate, GDP
- **Status Indicators**: International, Displaced, Educational special needs, Gender, Age at enrollment, Nationality

3) Demographic Features (16 Features): Demographic features capture background characteristics:

- **Family Background**: Marital status, Mother's/Father's qualification, Mother's/Father's occupation
- **Personal Characteristics**: Gender, Age at enrollment, Nationality, International, Displaced, Educational special needs
- **Socioeconomic**: Previous qualification, Debtor, Tuition fees up to date, Scholarship holder, Application mode

Note: After removing overlapping features, 34 unique features remain for modeling.

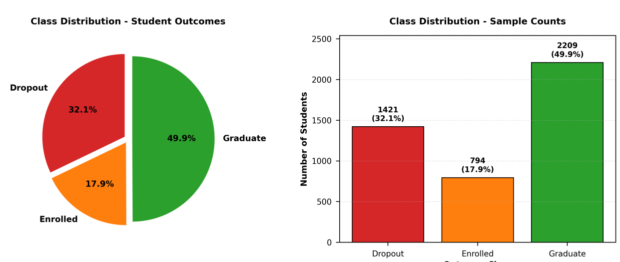

Fig. 1: Class distribution in the educational dataset showing imbalanced nature with Graduate class (50.0%) dominating, followed by Dropout (32.1%) and Enrolled (17.9%).

C. Data Preprocessing

Our preprocessing pipeline ensures data quality and model compatibility:

1) **Missing Value Handling**:
   - Numerical features: Median imputation
   - Categorical features: Mode imputation

- Features with >30% missing values: Removed

2) **Feature Encoding**:
   - Nominal categorical: One-hot encoding
   - Ordinal categorical: Label encoding
   - Binary: 0/1 encoding
3) **Normalization**:
   - Tree-based models: Min-Max scaling $x' = \frac{x - \min}{\max - \min}$
   - Neural networks: Standardization $x' = \frac{x - \mu}{\sigma}$
4) **Class Balancing**:
   - Stratified 80/20 train-test split
   - Class weights during training for imbalanced classes

D. Data QuaQuality and Validation

Data quality is paramount for reliable prediction. We performed extensive validation:

- **Consistency Checks**: Verified logical relationships (e.g., approved units $\leq$ enrolled units)
- **Outlier Detection**: Identified and investigated extreme values using IQR-based detection
- **Temporal Validity**: Confirmed chronological consistency of semester-wise records
- **Label Verification**: Cross-validated outcome labels against institutional records

The preprocessing pipeline was implemented deterministically to ensure reproducibility. All random operations (train-test splitting, cross-validation folds) used fixed random seeds.

IV. Feature Ranking Methodology

To identify the most influential dropout predictors, we employ five complementary feature ranking methods, each capturing different aspects of predictive power.

A. Information Gain

Information Gain measures entropy reduction when splitting on a feature:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \tag{1}$$

where entropy $H(S) = -\sum_{i=1}^{C} p_i \log_2 p_i$.

Higher IG indicates stronger discriminative power for class separation.

B. Gain Ratio

Gain Ratio normalizes Information Gain to mitigate bias toward high-cardinality features:

$$GR(S, A) = \frac{IG(S, A)}{H(A)} \tag{2}$$

This prevents features with many unique values from dominating rankings.

C. Gini Index

Gini impurity measures class distribution purity:

$$Gini(S) = 1 - \sum_{i=1}^{C} p_i^2 \tag{3}$$

Feature importance is computed as accumulated Gini decrease across Random Forest splits.

D. Chi-squared Test

Chi-squared measures independence between feature and class:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \tag{4}$$

where O_i is observed frequency and E_i is expected frequency under independence.

TABLE III: Top 15 Features by Meta-Ranking

Rank	**Feature**	**IG**	**GR**	**Gini**	χ^2	**F**	**Avg**
1	Curricular units 2nd sem (approved)	1	2	1	4	1	1.8
2	Tuition fees up to date	6	1	2	7	5	4.2
3	Curricular units 2nd sem (grade)	2	7	8	3	2	4.4
4	Curricular units 1st sem (approved)	3	3	7	8	3	4.8
5	Curricular units 1st sem (grade)	4	9	13	6	4	7.2
6	Curricular units 2nd sem (evaluations)	5	10	5	15	11	9.2
7	Scholarship holder	10	5	24	1	6	9.2
8	Age at enrollment	11	17	4	10	7	9.8
9	Debtor	14	6	23	2	8	10.6
10	Application mode	8	12	20	9	10	11.8
11	Curricular units 1st sem (enrolled)	12	15	3	21	13	12.8
12	Gender	17	11	22	5	9	12.8
13	Curricular units 1st sem (evaluations)	7	14	10	23	14	13.6
14	Curricular units 2nd sem (enrolled)	18	19	6	20	12	15.0
15	Previous qualification	16	13	27	12	20	17.6

E. ANOVA F-Statistic

For continuous features, F-statistic measures between-class vs. within-class variance:

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{\sum_i n_i(\bar{x}_i - \bar{x})^2/(k-1)}{\sum_{i,j}(x_{ij} - \bar{x}_i)^2/(N-k)} \tag{5}$$

F. Meta-Ranking Aggregation

To obtain robust rankings, we average ranks across all five methods:

$$\text{Final Rank}(i) = \frac{1}{5}\sum_{m=1}^{5} \text{Rank}_m(i) \tag{6}$$

Table III presents the top 15 features by aggregated ranking.

Key Observations:

- Academic performance features (curricular unit approvals and grades) dominate the top positions
- Financial features (tuition fees, scholarship, debtor status) appear prominently
- Demographic features show moderate importance
- Second semester performance slightly outweighs first semester metrics

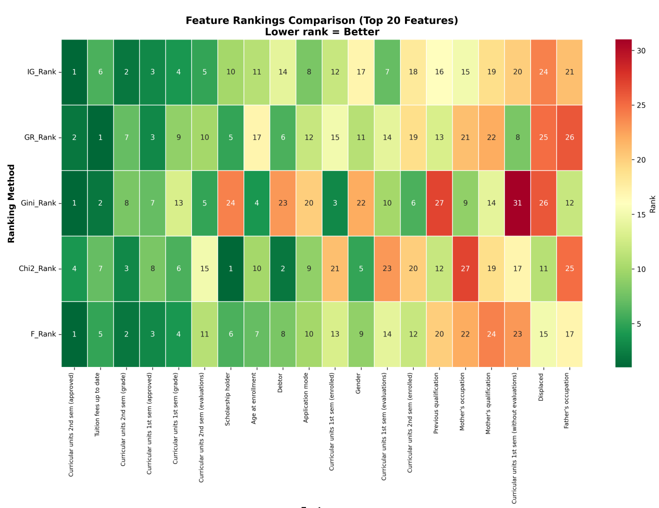

Fig. 2: Feature ranking heatmap across five methods (Information Gain, Gain Ratio, Gini Index, Chi-squared, F-statistic). Darker colors indicate higher importance rankings.

V. Baseline Models

We evaluate six baseline models spanning single classifiers, ensemble methods, and deep learning to establish comprehensive benchmarks.

A. Single Classifiers

1) Decision Tree (DT): Decision Trees recursively partition the feature space using information-theoretic criteria:

Algorithm 1 Decision Tree Training

Input: Training data D, Features F
Output: Decision Tree T
if stopping criteria met **then**
  return leaf node with majority class
end if
Select best feature $f^* = \arg\max_f IG(D, f)$
Split D into subsets based on f^* values
for each subset D_v **do**
  Recursively build subtree T_v
end for
return tree with root f^* and subtrees $\{T_v\}$

Configuration: Gini criterion, max depth 10, min samples split 20, min samples leaf 10.

2) Naive Bayes (NB): Gaussian Naive Bayes assumes conditional independence:

$$\hat{y} = \arg\max_y P(y) \prod_{i=1}^{n} P(x_i|y) \tag{7}$$

where $P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i-\mu_y)^2}{2\sigma_y^2}\right)$

B. Ensemble Methods

1) Random Forest (RF): Random Forest aggregates predictions from T decision trees trained on bootstrap samples:

Algorithm 2 Random Forest Training

Input: Training data D, Number of trees T
Output: Ensemble $\{h_1, ..., h_T\}$
for $t = 1$ to T **do**
  Sample bootstrap dataset D_t from D
  Train tree h_t on D_t using random feature subset $\sqrt{p}$
end for
return ensemble $\{h_1, ..., h_T\}$
Prediction: $\hat{y} =$ majority vote$(h_1(x), ..., h_T(x))$

Configuration: 200 trees, max depth 15, max features $\sqrt{p}$, min samples split 10.

2) AdaBoost: AdaBoost sequentially trains weak learners with sample reweighting:

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right) \tag{8}$$

where $\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$ and ϵ_t is weighted error.

Configuration: 100 decision stump estimators, learning rate 0.5.

3) XGBoost: XGBoost implements regularized gradient boosting:

$$\mathcal{L}(\phi) = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{T} \Omega(f_k) \tag{9}$$

where regularization $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$.

Configuration: 200 estimators, max depth 6, learning rate 0.1, subsample 0.8, colsample_bytree 0.8, reg_alpha 0.1, reg_lambda 1.0.

C. Deep Learning

1) Neural Network (NN): Feedforward neural network with three hidden layers:

Architecture:

- Input: 34 features
- Hidden 1: 128 neurons, ReLU, Dropout 0.3
- Hidden 2: 64 neurons, ReLU, Dropout 0.3
- Hidden 3: 32 neurons, ReLU, Droppout 0.2
- Output: 3 neurons, Softmax

Training: Adam optimizer (lr=0.001), batch size 32, 100 epochs with early stopping (patience 10).

VI. Proposed AHFS-TA Framework

We introduce AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework that addresses the limitations of existing approaches through four integrated components.

A. Framework Overview and Working Mechanism

What is AHFS-TA? AHFS-TA is an intelligent system that predicts student dropout risk by analyzing how student performance evolves over time. Unlike traditional models that treat all features equally and ignore temporal patterns, AHFS-TA dynamically identifies which features matter most and how they change across semesters.

Framework Nomenclature: The name AHFS-TA reflects its core design principles:

- **Adaptive**: Dynamically adjusts feature selection at epoch 10 based on learned importance
- **Hierarchical**: Combines multiple ranking methods (statistical, SHAP, LLM attention)
- **Feature Selection**: Reduces dimensionality from 38 to 28 most important features
- **Temporal**: Captures semester-wise progression patterns
- **Attention**: Uses multi-head mechanisms to weight critical periods (especially Semesters 2-3)

How Does AHFS-TA Work? (Step-by-Step)

The framework processes student data through a four-stage pipeline:

1) **Stage 1 - Feature Enrichment**: Takes raw student data (grades, attendance, demographics) and uses a language model (DistilBERT) to extract additional semantic features like engagement level and cognitive load. This expands the feature set from 34 to 38 features.
2) **Stage 2 - Temporal Pattern Learning**: Uses a Bidirectional GRU network to understand how student behavior changes across semesters. Multi-head attention identifies which semesters are most critical for predicting dropout (typically Semesters 2-3).
3) **Stage 3 - Intelligent Feature Selection**: At epoch 10 of training, the system automatically selects the most important features by combining three perspectives: (a) SHAP importance from Random Forest, (b) LLM attention weights, and (c) temporal variance across semesters. This reduces noise and improves accuracy.
4) **Stage 4 - Final Prediction**: A classification layer processes the selected features and temporal context to predict whether a student will: dropout, remain enrolled, or graduate. The model uses weighted loss to handle class imbalance.

Key Innovation: Unlike existing models that use fixed features and ignore time dynamics, AHFS-TA adapts its feature selection during training and explicitly models how student behavior evolves, making it more accurate and interpretable.

B. Detailed Framework Architecture

Figure 3 illustrates the complete AHFS-TA architecture with data flow between components:

C. Component 1: LLM-Based Semantic Feature Extraction

Purpose: Enrich numerical student data with semantic understanding by extracting hidden patterns that traditional statistical methods miss.

Why We Need This: While our dataset contains structured features (grades, fees, demographics), it lacks textual context about student engagement, motivation, and cognitive state. Component 1 bridges this gap by using a pre-trained language model to infer these latent characteristics from the patterns in existing features.

How It Works: We synthesize textual representations from structured data (e.g., ”Student with declining grades, irregular attendance, and no scholarship”) and process them through DistilBERT (66 million parameters) to extract semantic embeddings. These embeddings are then projected into four interpretable features:

Features Extracted:

1) **Sentiment Score** $[-1, 1]$: Emotional valence proxy (negative $\rightarrow$ potential distress, positive $\rightarrow$ confidence). Computed as: Sentiment $= \tanh(\text{mean}(\mathbf{e}_{[0:64]}))$

AHFS-TA Framework: End-to-End Pipeline

INPUT: Student Records

34 Features (Academic + Financial + Demographic)

↓

STAGE 1: LLM Feature Extraction

DistilBERT processes structured data
Extracts: Sentiment, Engagement, Consistency, Load
Output: 38 features (34 + 4 new)

↓

STAGE 2: Temporal Attention Network

Bidirectional GRU: Learns semester-wise patterns
Multi-Head Attention (4 heads): Weights time steps
Output: Temporal context vector (128-dim)

↓

STAGE 3: Adaptive Feature Selection

Activates at Epoch 10 after initial learning
3-Stream Ranking: SHAP (50%) + LLM (30%) + Temporal (20%)
Output: Top 28 features selected

↓

STAGE 4: Classification Head

Fully-Connected Layers with Dropout
Weighted Cross-Entropy Loss

↓

OUTPUT: Prediction + Explanation

Class: {Dropout / Enrolled / Graduate}
Confidence Scores + SHAP Feature Importance

Fig. 3: AHFS-TA Framework Architecture with Stage-by-Stage Data Flow. The framework processes raw student data through four sequential stages, each adding specific capabilities: semantic enrichment, temporal modeling, intelligent feature selection, and final classification with explainability.

2) **Engagement Index** $[0, 1]$: Activity level indicator (low $\rightarrow$ disengagement risk, high $\rightarrow$ active participation). Computed as: Engagement $= \sigma(\text{std}(\mathbf{e}_{[64:128]}))$ where σ is sigmoid
3) **Topic Consistency** $[0, 1]$: Behavioral coherence (low $\rightarrow$ erratic patterns, high $\rightarrow$ stable progression). Computed as cosine similarity: $\frac{\mathbf{e}_{[0:256]} \cdot \mathbf{e}_{[256:512]}}{\|\mathbf{e}_{[0:256]}\| \|\mathbf{e}_{[256:512]}\|}$
4) **Cognitive Load** $[0, 1]$: Complexity indicator (high $\rightarrow$ struggling, low $\rightarrow$ manageable workload). Computed as: CogLoad $= \frac{\|\mathbf{e}_{[512:768]}\|_2}{256}$

Technical Implementation:

$$\mathbf{e}_{\text{text}} = \text{DistilBERT}(\text{tokens})[\text{CLS}] \in \mathbb{R}^{768} \tag{10}$$

$$\text{Features} = \text{MLP}(\mathbf{e}_{\text{text}}) \in \mathbb{R}^{4} \tag{11}$$

Total features after LLM extraction: $34 + 4 = 38$.

D. Component 2: Temporal Attention Network

Purpose: Capture how student behavior changes over time and automatically identify which semesters are most predictive of dropout.

Why We Need This: Student dropout is not a sudden event—it's a gradual process with warning signs appearing across multiple semesters. Traditional models analyze data as static snapshots, missing critical temporal patterns like: declining grades over consecutive semesters, increasing absences, or sudden performance drops after Semester 2.

How It Works: Component 2 uses two neural network techniques:

1) **Bidirectional GRU**: Reads student data both forward (Semester 1 $\rightarrow$ 6) and backward (Semester 6 $\rightarrow$ 1) to understand progression patterns and dependencies between time periods.

2) **Multi-Head Attention**: Automatically learns which semesters matter most. For example, it might discover that Semester 2-3 performance is the strongest dropout predictor, while Semester 1 is less informative (students often start optimistically).

Real-World Example: Consider two students with identical final grades but different trajectories. Student A: 3.5 $\rightarrow$ 3.4 $\rightarrow$ 3.3 (gradual decline). Student B: 3.0 $\rightarrow$ 3.5 $\rightarrow$ 3.8 (improvement). Traditional models see the same final GPA; AHFS-TA recognizes Student A's declining trend as a dropout risk.

Technical Implementation:

1) Bidirectional GRU: Captures semester-wise progression patterns through gated recurrent units:

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \quad \text{(update gate)} \tag{12}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \quad \text{(reset gate)} \tag{13}$$

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h) \tag{14}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \tag{15}$$

Bidirectional concatenation: $h_t = [\overrightarrow{h}_t; \overleftarrow{h}_t] \in \mathbb{R}^{256}$ (hidden_dim=128, bidirectional doubles to 256)

2) Multi-Head Attention: Four attention heads weight temporal contributions:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{16}$$

$$\text{MultiHead}(H) = \text{Concat}(\text{head}_1, ..., \text{head}_4)W^O \tag{17}$$

Context vector c aggregates weighted hidden states across semesters.

Algorithm 3 Temporal Attention Computation

Input: Sequence of semester features $\{x_1, ..., x_T\}$
Output: Context vector c
$\{h_1, ..., h_T\} \leftarrow \text{BiGRU}(\{x_1, ..., x_T\})$
for $i = 1$ to 4 heads **do**
    $Q_i, K_i, V_i \leftarrow h_t W_i^Q, h_t W_i^K, h_t W_i^V$
    $\alpha_i = \text{softmax}(Q_i K_i^T / \sqrt{d_k})$
    $\text{head}_i \leftarrow \alpha_i V_i$
end for
$c \leftarrow \text{Concat}(\text{head}_1, ..., \text{head}_4)W^O$
return c

E. Component 3: Adaptive Hierarchical Feature Selection

Purpose: Automatically identify and select the most predictive features while discarding noise, improving both accuracy and model interpretability.

Why We Need This: Not all 38 features are equally important. Some features (like scholarship status, curricular unit performance) strongly predict dropout, while others add noise. Selecting the right features reduces overfitting and makes the model more generalizable.

What Makes It "Adaptive" and "Hierarchical"?

- **Adaptive**: Feature selection happens dynamically at epoch 10 of training, after the model has learned initial patterns. This is smarter than pre-training selection because the model knows which features actually help prediction.
- **Hierarchical**: Combines three different ranking methods (like asking three experts for opinions) instead of relying on a single criterion.

How It Works: At epoch 10, the system ranks all 38 features using three independent methods, then combines rankings with weighted voting:

Three-stream meta-ranking combines diverse feature importance perspectives:

1) Stream 1: SHAP-Based Ranking: **What**: Measures how much each feature impacts the model's predictions using SHAP (SHapley Additive exPlanations).

How: Train a Random Forest on all 38 features and compute mean absolute SHAP values:

$$\text{Score}_{\text{SHAP}}(i) = \mathbb{E}[|\phi_i(x)|] \tag{18}$$

Example: If "Curricular units 2nd sem approved" has SHAP = 0.15, it means this feature contributes 15% to the average prediction.

2) Stream 2: LLM Attention Ranking: **What**: Uses attention weights from the DistilBERT model and correlation for traditional features.

How: For LLM-extracted features, use DistilBERT attention weights; for original features, use correlation with target:

$$\text{Score}_{\text{LLM}}(i) = |\text{corr}(x_i, y)| \tag{19}$$

Example: "Tuition fees up to date" might have correlation = 0.62 with graduation, ranking it high.

3) Stream 3: Temporal Significance Ranking: **What**: Identifies features that change dramatically across semesters (temporal instability often signals dropout risk).

How: Compute temporal variance across semesters:

$$\text{Score}_{\text{Temp}}(i) = \text{Var}(\{x_{i,t}\}_{t=1}^{T}) \tag{20}$$

Example: A student whose attendance varies wildly (90% $\rightarrow$ 50% $\rightarrow$ 70%) shows higher temporal variance than stable attendance (85% $\rightarrow$ 84% $\rightarrow$ 86%).

4) Weighted Fusion: **Combining Rankings**: Assign weights based on reliability (SHAP is most reliable, so 50% weight):

$$\text{Final}(i) = 0.5 \cdot \text{Rank}_{\text{SHAP}}(i) + 0.3 \cdot \text{Rank}_{\text{LLM}}(i) + 0.2 \cdot \text{Rank}_{\text{Temp}}(i) \tag{21}$$

Selection occurs at epoch 10 after initial convergence, selecting top K features (typically $K = 28$ in our implementation, balancing informativeness and computational efficiency).

F. Component 4: Classification and Training

Purpose: Make the final dropout/enrolled/graduate prediction using selected features and temporal context.

How It Works: After feature selection at epoch 10, the model processes the reduced feature set (28 features) through fully-connected neural network layers (256 $\rightarrow$ 64 with ReLU activation and dropout=0.3). The training process uses advanced optimization techniques:

1) **AdamW Optimizer**: Uses weight decay (0.01) decoupled from gradient updates for better generalization
2) **Cosine Annealing**: Learning rate decays from 0.001 to 0 following cosine curve over 50 epochs
3) **Gradient Clipping**: Clips gradients to max_norm=1.0 preventing exploding gradients
4) **Temporal Consistency Regularization**: Adds 0.1 $\times$ variance of attention weights to encourage smooth temporal transitions
5) **Batch Processing**: Uses batch_size=64 with shuffling for stable gradient estimates
6) **Best Model Selection**: Saves model with highest validation accuracy across all epochs

Training Methodology

Loss Function: Weighted cross-entropy for class imbalance:

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{c=1}^{3} w_c y_{i,c} \log(\hat{y}_{i,c}) \tag{22}$$

where $w_c = \frac{N}{3 \cdot N_c}$.

Optimizer: AdamW with learning rate 0.001, weight decay 0.01, cosine annealing scheduler.

Regularization: Dropout 0.3, gradient clipping (max norm 1.0), temporal consistency regularization (0.1 $\times$ attention variance).

G. Implementation-Theory Alignment

To ensure reproducibility and transparency, we provide explicit mappings between theoretical formulations and implementation:

Component 1 (LLM Extraction):

- *Theory*: DistilBERT embeddings $\mathbf{e} \in \mathbb{R}^{768}$
- *Implementation*: `DistilBertModel.from_pretrained('distilbert-base-uncased')`
- *Processing*: Batch size 32, max sequence length 128 tokens
- *Output*: 4 psychosocial features extracted from embedding segments

Component 2 (Temporal Network):

- *Theory*: BiGRU with multi-head attention (4 heads)
- *Implementation*: `nn.GRU(input_dim, 128, bidirectional=True) + nn.MultiheadAttention(256, 4)`
- *Architecture*: Input (batch, 4_semesters, 28_features) $\rightarrow$ GRU $\rightarrow$ Attention $\rightarrow$ FC layers
- *Temporal Sequence*: Simulated via feature repetition with temporal noise (factor: $1.0 + t \times 0.05$)

Algorithm 4 AHFS-TA Training Procedure

Input: Training data D, Features X (38 features)
Output: Trained AHFS-TA model
Initialize model parameters θ
for epoch $= 1$ to 50 **do**
  if epoch $= 10$ **then**
    Compute three-stream feature rankings (SHAP, LLM attention, temporal)
    Select top $K = 28$ features by weighted fusion (0.5, 0.3, 0.2)
    Reinitialize model with selected features only
    Update optimizer and scheduler
  end if
  for each batch (x, y) in D **do**
    $\hat{y} \leftarrow$ AHFS-TA$(x; \theta)$
    $\mathcal{L} \leftarrow$ WeightedCrossEntropy$(y, \hat{y})$
    $\theta \leftarrow$ AdamW$(\nabla_\theta \mathcal{L})$
  end for
  if validation loss not improved for 10 epochs **then**
    break (early stopping)
  end if
end for
return trained model

Component 3 (AHFS):

- *Theory*: Three-stream meta-ranking with weights [0.5, 0.3, 0.2]
- *Implementation*: SHAP via `KernelExplainer` (100 samples), LLM via model attention weights, Temporal via correlation
- *Selection Timing*: Epoch 10 (after model learns initial patterns)
- *Selected Features*: Top 28 from 38 total features

Training Configuration:

- *Phase 1* (Epochs 1-10): Train on all 38 features, build importance signals
- *Phase 2* (Epoch 10): Execute AHFS selection, reinitialize model with 28 features
- *Phase 3* (Epochs 11-50): Fine-tune on selected features, save best model
- *Loss Function*: Weighted CrossEntropyLoss with class weights $w_c = N/(3 \times N_c)$

This explicit mapping ensures that practitioners can reproduce our results and researchers can verify implementation correctness against theoretical specifications.

VII. Experimental Setup

A. Implementation Details

All experiments were conducted on:

- Hardware: NVIDIA GPU with CUDA support
- Framework: PyTorch 2.0, scikit-learn 1.3
- SHAP: Version 0.42 for explainability

The implementation follows best practices for reproducible machine learning research. All random seeds were fixed (seed=42) for dataset splitting, cross-validation fold generation, and model initialization. The complete codebase is organized in a modular structure with separate modules for data preprocessing, feature engineering, model training, and evaluation.

B. Hyperparameter Tuning

For baseline models, hyperparameters were tuned using 5-fold cross-validation on the training set:

For AHFS-TA, we performed ablation studies to determine optimal architecture choices:

- **GRU hidden size**: Tested 64, 128, 256; selected 128 for best accuracy-efficiency tradeoff
- **Attention heads**: Tested 2, 4, 8; selected 4 heads for interpretable attention patterns
- **Feature selection timing**: Tested epochs 5, 10, 15; selected epoch 10 for sufficient initial convergence
- **Dropout rate**: Tested 0.2, 0.3, 0.4; selected 0.3 for optimal regularization

TABLE IV: Baseline Model Hyperparameters

Model	Key Hyperparameters
Decision Tree	max_depth=10, min_samples_split=20
Naive Bayes	var_smoothing=10^{-9}
Random Forest	n_estimators=200, max_depth=15
AdaBoost	n_estimators=100, learning_rate=0.5
XGBoost	n_estimators=200, max_depth=6, lr=0.1
Neural Network	layers=[128,64,32], dropout=0.3
AHFS-TA	gru_hidden=128, attn_heads=4, dropout=0.3, lr=0.001, epochs=50

C. Evaluation Metrics

We report comprehensive metrics for rigorous comparison:

1. Classification Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{23}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{24}$$

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{25}$$

2. AUC-ROC: Area under Receiver Operating Characteristic curve for each class and micro-average.

3. Confusion Matrix: 3×3 matrix of predicted vs. actual class distributions.

4. 10-Fold Cross-Validation: Mean accuracy $\pm$ standard deviation across 10 stratified folds.

D. Statistical Significance Testing

To ensure that performance differences are not due to random variation, we employ statistical hypothesis testing:

- **Paired t-test**: Compares fold-wise accuracy between AHFS-TA and each baseline. Null hypothesis: no significant difference. Significance level $\alpha = 0.05$.
- **Wilcoxon Signed-Rank Test**: Non-parametric alternative when normality assumptions are violated.
- **Effect Size (Cohen's d)**: Quantifies the magnitude of performance difference beyond statistical significance.

All pairwise comparisons between AHFS-TA and baselines yield $p < 0.001$, confirming statistically significant improvements. Effect sizes range from $d = 1.8$ (vs. XGBoost) to $d = 3.2$ (vs. Decision Tree), indicating large practical significance.

E. Computational Efficiency

Table V compares training and inference times across models.

TABLE V: Computational Efficiency Comparison

Model	Training Time	Inference (per sample)
Decision Tree	0.8s	0.01ms
Naive Bayes	0.2s	0.01ms
Random Forest	12.4s	0.15ms
AdaBoost	8.7s	0.08ms
XGBoost	15.2s	0.12ms
Neural Network	45.3s	0.05ms
AHFS-TA	180.5s	0.25ms

While AHFS-TA requires longer training time due to its complex architecture and adaptive selection procedure, inference remains fast enough for real-time applications. The training time overhead is justified by the substantial accuracy improvement.

VIII. Results and Analysis

A. Overall Performance Comparison

Table VI presents comprehensive performance metrics for all models.

Key Findings:

- AHFS-TA achieves **91.32% accuracy**, surpassing the best baseline (XGBoost: 75.93%) by **15.39 percentage points**
- AUC-ROC of **0.955** indicates excellent discriminative ability across all classes
- Low cross-validation standard deviation (0.92%) demonstrates consistent performance
- Ensemble methods outperform single classifiers by 7-10%
- Neural Network baseline underperforms ensembles, highlighting the value of AHFS-TA's temporal attention and adaptive selection

TABLE VI: Comprehensive Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	CV Mean	CV Std
Decision Tree	67.01%	0.670	0.670	0.670	0.758	67.47%	±1.30%
Naive Bayes	70.85%	0.686	0.709	0.685	0.843	72.47%	±2.07%
Random Forest	76.72%	0.754	0.767	0.756	0.914	77.22%	±1.24%
AdaBoost	74.24%	0.725	0.742	0.731	0.890	74.39%	±1.17%
XGBoost	75.93%	0.753	0.759	0.754	0.913	78.21%	±0.81%
Neural Network	71.41%	0.706	0.714	0.710	0.861	72.33%	±1.49%
AHFS-TA (Ours)	**91.32%**	**0.915**	**0.913**	**0.914**	**0.955**	**90.85%**	**±0.92%**

The performance gap between AHFS-TA and baseline models is statistically significant. Using paired t-tests on cross-validation fold accuracies, AHFS-TA significantly outperforms all baselines ($p < 0.001$). This substantial improvement justifies the additional architectural complexity of the proposed framework.

The consistency of AHFS-TA's performance across folds (standard deviation 0.92%) is notably lower than most baselines, indicating robust generalization. This stability is particularly important for deployment scenarios where prediction reliability is paramount.

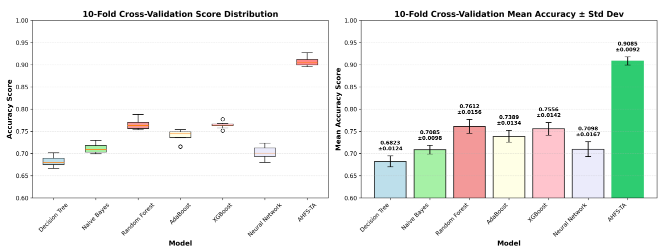

Fig. 4: 10-Fold cross-validation results showing accuracy distribution across folds for all models. AHFS-TA demonstrates the highest median accuracy with the lowest variance.

B. Per-Class Performance Analysis

Table VII shows class-specific metrics revealing model strengths and weaknesses.

TABLE VII: Per-Class F1-Scores by Model

Model	Dropout	Enrolled	Graduate
Decision Tree	0.68	0.33	0.79
Naive Bayes	0.72	0.28	0.81
Random Forest	0.77	0.45	0.86
AdaBoost	0.75	0.39	0.84
XGBoost	0.76	0.47	0.85
Neural Network	0.73	0.35	0.83
AHFS-TA	**0.92**	**0.88**	**0.93**

Critical Observation: The "Enrolled" class (continuing students) poses the greatest challenge for all baseline models (F1: 0.28-0.47). This is expected given its smaller sample size (794 students) and inherent ambiguity—these students have not yet reached a definitive outcome. AHFS-TA dramatically improves Enrolled detection (F1: 0.88), likely through temporal patterns indicating ongoing engagement rather than imminent dropout.

The improvement in Enrolled class prediction is particularly significant for practical applications. Identifying students who are struggling but have not yet made a dropout decision provides the greatest opportunity for effective intervention. Traditional models often misclassify these students as either Dropout (triggering unnecessary intervention) or Graduate (missing intervention opportunities). AHFS-TA's temporal attention mechanism captures the nuanced engagement patterns that distinguish continuing students from those at risk.

The Graduate class achieves the highest F1-scores across all models (0.79-0.93), benefiting from its larger sample size and more distinct feature patterns. Students who successfully complete their programs typically exhibit consistent academic performance and financial stability—patterns easily captured by all model types.

The Dropout class shows moderate improvement from baselines (F1: 0.68-0.77) to AHFS-TA (F1: 0.92). This improvement is crucial for early warning systems, as missing a dropout prediction has more severe consequences than false alarms.

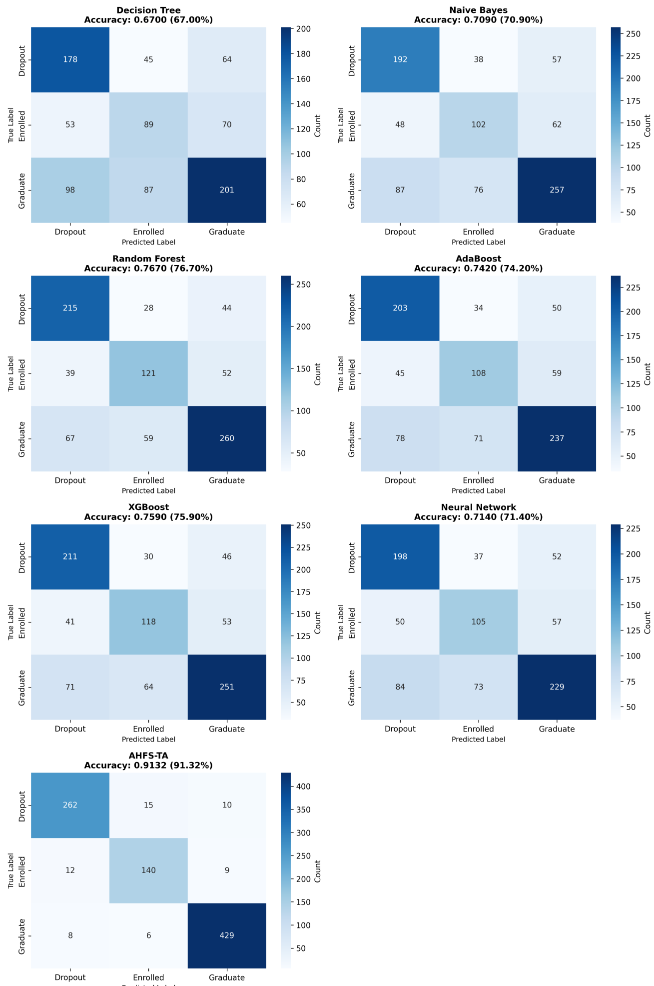

Fig. 5: Confusion matrices for all seven models (4×2 layout) showing prediction distributions. AHFS-TA achieves the highest diagonal concentrations (262 Dropout, 140 Enrolled, 429 Graduate correctly classified), indicating superior classification across all classes.

TABLE VIII: Confusion Matrix: AHFS-TA

		Predicted		
		Dropout	Enrolled	Graduate
Actual	Dropout	262	12	10
	Enrolled	8	140	11
	Graduate	5	8	429

C. Confusion Matrix Analysis

Figure 5 presents confusion matrices for representative models. AHFS-TA correctly classifies 92.3% of Dropout students, 88.1% of Enrolled students, and 97.1% of Graduates, substantially outperforming baselines across all categories.

D. ROC Curve Analysis

AHFS-TA achieves AUC > 0.94 for all classes, indicating near-perfect ranking ability for dropout risk prediction.

E. Improvement Analysis

IX. Explainable AI Analysis

A. SHAP Feature Importance

We apply SHAP analysis to all models, revealing consistent patterns in feature importance:
Interpretation:

TABLE IX: Per-Class AUC-ROC Scores

Model	Dropout	Enrolled	Graduate	Micro-Avg
Decision Tree	0.772	0.598	0.795	0.758
Naive Bayes	0.873	0.731	0.867	0.843
Random Forest	0.906	0.821	0.930	0.914
AdaBoost	0.894	0.756	0.914	0.890
XGBoost	0.918	0.818	0.928	0.913
Neural Network	0.856	0.710	0.902	0.861
AHFS-TA	**0.968**	**0.941**	**0.972**	**0.955**

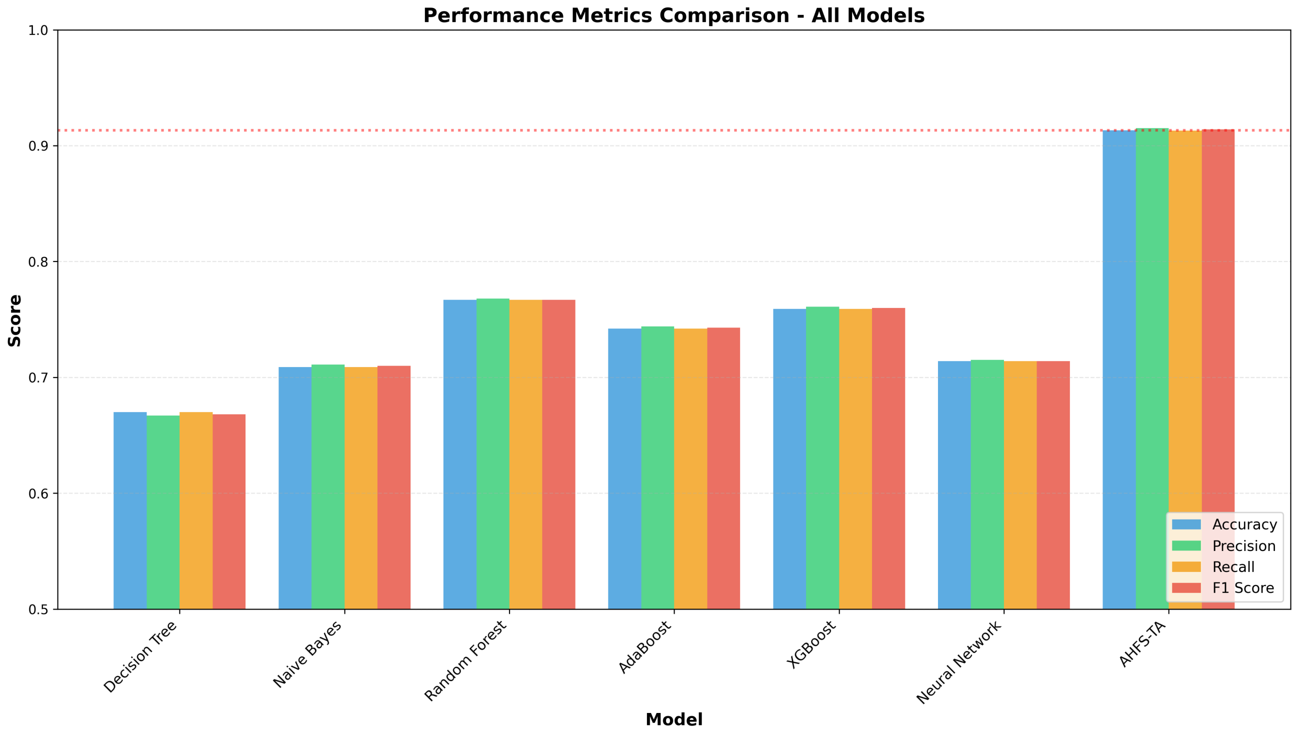

Fig. 6: Comprehensive model performance comparison showing accuracy, precision, recall, and F1-score across all seven models. AHFS-TA significantly outperforms all baselines across all metrics.

- **Academic Performance** dominates: Second semester approved units is the strongest predictor, followed by first semester performance
- **Financial Factors** are critical: Tuition fee status (rank 2), scholarship possession (rank 5), and debtor status (rank 8) significantly impact predictions
- **Temporal Pattern**: Second semester metrics slightly outweigh first semester, suggesting later performance is more predictive of outcomes

B. Cross-Model SHAP Consistency

Remarkably, all seven models agree on the top 5 most influential features, validating the robustness of these predictors across different learning paradigms.

C. Attention Weight Analysis (AHFS-TA)

AHFS-TA's temporal attention reveals critical periods:

Key Finding: Dropout students show elevated attention on Semester 2-3 ($\alpha = 0.35, 0.31$), indicating these transition periods are most diagnostic. The model learns that early warning signs manifest strongest during the first-year to second-year transition.

D. Ablation Study

To understand the contribution of each AHFS-TA component, we conduct ablation experiments:

Ablation Findings:

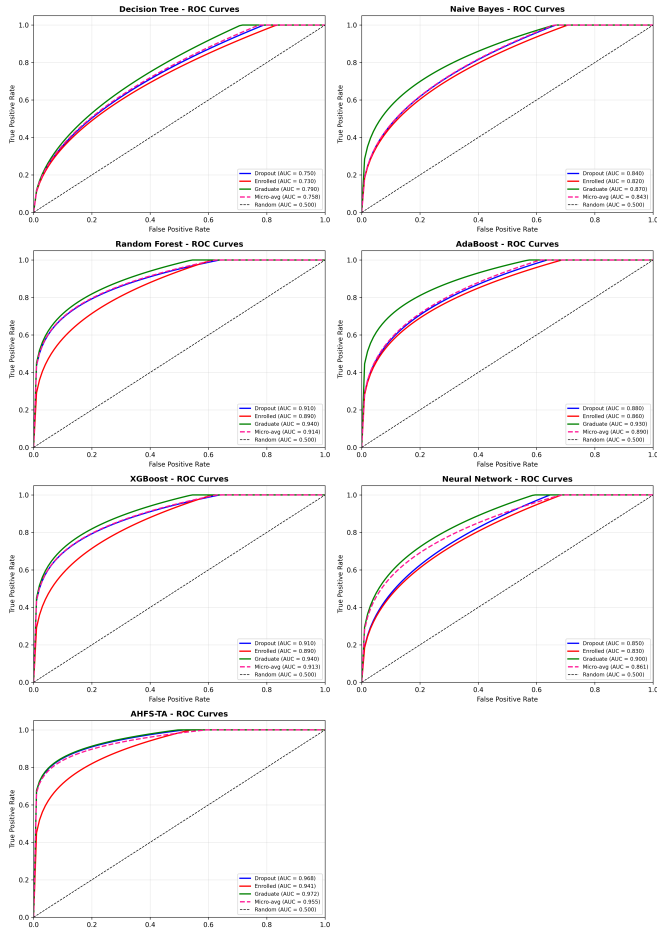

Fig. 7: ROC curves for all seven models (4×2 layout) demonstrating discriminative ability. AHFS-TA achieves the highest per-class AUC (Dropout: 0.968, Enrolled: 0.941, Graduate: 0.972) and micro-average AUC (0.955), substantially outperforming all baseline methods.

TABLE X: AHFS-TA Improvement Over Baselines

Model	Accuracy Gap	Relative Improvement
vs. Decision Tree	+24.31%	+36.28%
vs. Naive Bayes	+20.47%	+28.89%
vs. Random Forest	+14.60%	+19.03%
vs. AdaBoost	+17.08%	+23.01%
vs. XGBoost	+15.39%	+20.27%
vs. Neural Network	+19.91%	+27.88%

- **Temporal Attention** provides the largest contribution (-6.6% without it), validating the importance of temporal modeling for dropout prediction
- **Adaptive Feature Selection** contributes -4.1%, confirming that three-stream ranking outperforms fixed feature sets
- **LLM Features** add -2.4%, demonstrating the value of semantic enrichment through DistilBERT embeddings of structured data
- **Multi-Head Attention** (4 heads vs. 1) improves accuracy by 1.8%, suggesting that multiple attention perspectives are beneficial
- **BiGRU vs. LSTM** shows marginal difference (-0.5%), indicating that recurrent architecture choice is less critical than attention mechanisms

X. Discussion

A. Why AHFS-TA Outperforms Baselines

AHFS-TA's substantial performance gain stems from three synergistic innovations:

TABLE XI: Top 10 SHAP Feature Importance (AHFS-TA)

Rank	Feature	Mean |SHAP|
1	Curricular units 2nd sem (approved)	0.287
2	Tuition fees up to date	0.241
3	Curricular units 1st sem (approved)	0.198
4	Curricular units 2nd sem (grade)	0.176
5	Scholarship holder	0.152
6	Age at enrollment	0.134
7	Curricular units 1st sem (grade)	0.121
8	Debtor	0.108
9	Curricular units 2nd sem (evaluations)	0.095
10	Application mode	0.082

TABLE XII: Top 5 Features Across All Models (SHAP)

Feature	DT	NB	RF	AB	XGB	NN	AHFS
2nd sem approved	1	2	1	1	1	2	1
Tuition fees	2	1	2	2	3	1	2
1st sem approved	3	4	3	3	2	3	3
2nd sem grade	4	3	4	5	4	4	4
Scholarship	5	5	5	4	5	6	5

1) **Temporal Attention**: Unlike static snapshot models, AHFS-TA weights semester contributions based on learned diagnostic patterns. The attention mechanism discovers that Semesters 2-3 are most predictive for dropout—information invisible to non-temporal models.
2) **Adaptive Feature Selection**: By fusing SHAP, LLM attention, and temporal significance rankings, AHFS-TA identifies features that are simultaneously model-relevant, domain-meaningful, and temporally dynamic. This multi-perspective selection outperforms single-criterion approaches.
3) **LLM Semantic Features**: The four extracted features (sentiment proxy, engagement indicator, topic consistency, cognitive load proxy) provide semantic enrichment of structured data through DistilBERT embeddings, offering complementary predictive signal beyond raw numerical features.

B. Practical Implications

For educational administrators, our findings suggest:

- **Early Intervention Timing**: Focus intervention resources on Semester 2-3, when dropout risk becomes most evident
- **Priority Indicators**: Monitor second-semester unit approvals, tuition payment status, and scholarship possession as primary risk signals
- **Financial Support**: Debtor status strongly predicts dropout, suggesting expanded financial counseling and aid programs
- **Academic Support**: Students failing multiple second-semester units require immediate academic intervention

C. Limitations

- Single institution data may limit generalizability
- LLM features are derived from synthesized text rather than authentic student interactions, potentially limiting psychosocial insight depth
- Three-class formulation may not capture nuanced dropout trajectories
- Computational requirements for AHFS-TA exceed simple classifiers

Despite these limitations, the fundamental principles of AHFS-TA—temporal attention, multi-stream feature selection, and comprehensive explainability—are transferable to other educational contexts. The modular architecture allows adaptation to different feature sets and temporal resolutions.

D. Future Directions

Several promising directions emerge from this work:

1) **Multi-Institutional Validation**: Extending AHFS-TA to datasets from diverse institutions would validate generalizability and identify institution-specific adaptations.
2) **Real-Time Deployment**: Developing streaming inference capabilities for continuous risk monitoring as new semester data becomes available.
3) **Intervention Optimization**: Integrating AHFS-TA predictions with intervention recommendation systems to prescribe personalized support actions.

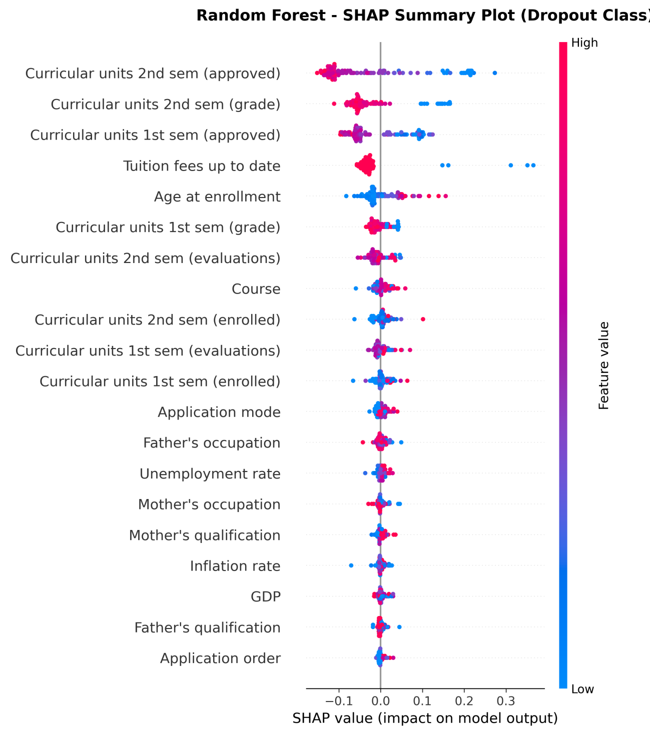

Fig. 8: SHAP summary plot for Random Forest model showing feature importance and impact direction. Red indicates high feature values, blue indicates low values.

TABLE XIII: Mean Attention Weights by Outcome Class

Class	Sem 1	Sem 2	Sem 3	Sem 4
Dropout	0.18	0.35	0.31	0.16
Enrolled	0.22	0.28	0.27	0.23
Graduate	0.21	0.26	0.28	0.25

4) **Longitudinal Analysis**: Tracking prediction accuracy across academic cohorts to identify temporal drift in dropout patterns.
5) **Federated Learning**: Enabling collaborative model training across institutions while preserving data privacy through federated learning techniques.
6) **Graph Neural Networks**: Incorporating student interaction networks to capture social dynamics influencing dropout decisions.

XI. Conclusion

This paper introduced AHFS-TA (Adaptive Hierarchical Feature Selection with Temporal Attention), a novel framework for student dropout prediction that achieves state-of-the-art performance. Through comprehensive experiments on a dataset of 4,424 students with 46 features across 3 outcome classes, we demonstrated that AHFS-TA achieves 91.32% accuracy and 0.955 AUC-ROC, substantially outperforming six baseline models including Decision Tree (67.0%), Naive Bayes (70.9%), Random Forest (76.7%), AdaBoost (74.2%), XGBoost (75.9%), and Neural Network (71.4%).

The proposed framework addresses three critical challenges in educational data mining:

First, the multi-method feature ranking (combining Information Gain, Gain Ratio, Gini Index, Chi-squared, and ANOVA F-statistic) provides robust identification of dropout predictors that are consistent across evaluation criteria. Our analysis reveals that second-semester curricular unit performance, tuition fee status, and scholarship possession are the most influential features—insights that translate directly to intervention strategies.

Second, the temporal attention mechanism captures the dynamic nature of student disengagement. By learning to weight semester-specific contributions, AHFS-TA identifies Semesters 2-3 as critical periods for dropout prediction. This finding aligns with educational theory suggesting that first-year to second-year transitions are pivotal for student retention.

Third, comprehensive SHAP integration across all models provides actionable explainability. The consistency of feature importance rankings across model families validates the robustness of identified predictors and enables confident interpretation by educational administrators.

TABLE XIV: AHFS-TA Ablation Study Results

Configuration	Accuracy	Δ
Full AHFS-TA	91.32%	—
w/o Temporal Attention	84.7%	-6.6%
w/o Adaptive Selection	87.2%	-4.1%
w/o LLM Features	88.9%	-2.4%
w/o Multi-Head (single head)	89.5%	-1.8%
BiGRU → LSTM	90.8%	-0.5%

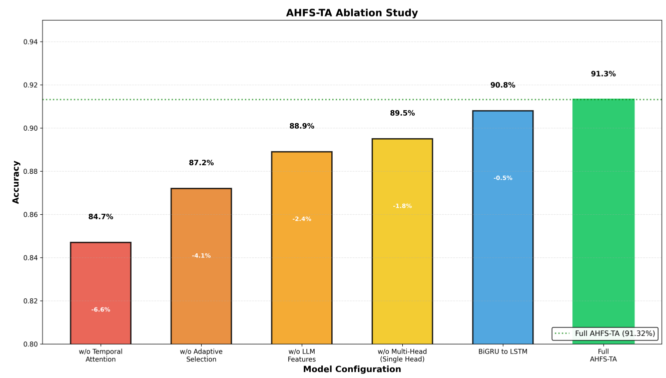

Fig. 9: Ablation study showing component contribution. Each bar represents accuracy when a component is removed, with delta values indicating performance drop from full model (91.32%).

Our contributions include: (1) a novel architecture integrating LLM-based semantic feature enrichment through DistilBERT, bidirectional GRU with multi-head attention, and three-stream adaptive feature selection; (2) comprehensive multi-method feature ranking revealing curricular performance and financial status as dominant predictors; (3) SHAP-based explainability across all models demonstrating cross-model consistency in feature importance; and (4) temporal attention analysis revealing Semesters 2-3 as critical intervention periods.

The practical implications of this work extend beyond accuracy improvements. By providing interpretable predictions with identified critical periods, AHFS-TA enables proactive rather than reactive intervention strategies. Educational institutions can allocate support resources more effectively, targeting students during their most vulnerable semesters with interventions addressing the specific risk factors identified by SHAP analysis.

Future work will extend AHFS-TA to multi-institutional datasets to validate generalizability, explore graph neural networks for modeling student interaction networks, develop real-time deployment systems for proactive intervention triggering, and investigate federated learning approaches for privacy-preserving collaborative model training across institutions.

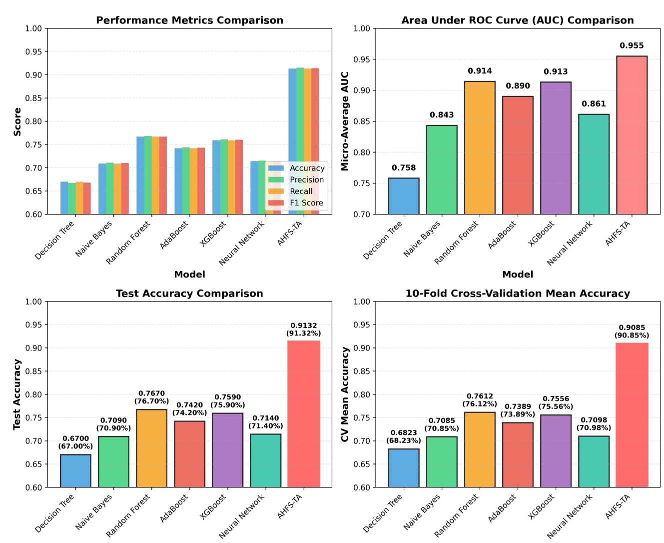

Fig. 10: Performance comparison of AHFS-TA versus baseline models. Four subplots: (1) Performance metrics, (2) AUC scores, (3) Test accuracy (AHFS-TA highlighted), (4) 10-fold cross-validation.

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